

ActInf Livestream #030.1 ~ “How to count biological minds: symbiosis, the free energy principle...”

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hello and welcome everybody welcome to the active inference lab today it's october 1st 5th sorry greeting shannon and 2021 we're going to be discussing this awesome paper how to count biological minds symbiosis the free energy principle and reciprocal multi-scale integration welcome to the active inference lab everyone we're a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links on this slide this is recorded in an archived live stream so please provide feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams like muting if there's noise in the background and raising our hands so we can all speak at this short link you'll find the calendar of past and future live streams the first tab is the regular tuesday live stream with two weeks per paper with the dot one and the dot two but then we have a few other series of streams related to different topics so go to that link to check out what's upcoming and see if you want to participate in one of these streams today in number 30.1 the goal is to learn and discuss this awesome paper on counting biological minds and we're really honored and appreciative that matt sims is here to discuss it with us so we're just going to introduce ourselves and walk through any of the figures any of the formalisms we also have a few other ideas and questions that lab participants have raised so this should be a great discussion thanks everyone for joining we'll just go around and say hi and introduce ourselves either just an introduction or feel free to say something that you are excited about with reading the paper and then we'll uh all non-authors will go and then we'll let matt just provide a little context here so i'm daniel i'm a postdoc researcher in california and i think i'm looking forward to drawing out some of these threads about collective behavior and the timeless questions about collective behavior with some of the new approaches that are discussed in this paper so i'll pass it to shannon hi there nice to see everyone again um i'm shannon i'm a phd candidate at the university of california in merced um and i've blurred my background but there is a dog rolling around so hopefully

it's not too distracting um i will pass it to stephen hello i'm stephen sillett i'm based in toronto um i'm doing a practice-based phd through the uk and around sort of psychological perspectives on practices particularly around social topographies and spatial meaning making in participatory theater and the arts and i'm really interested in this paper i like the folk psychology questions that it brings up and i like this idea of a folk biology which i found really interesting and the way that we have assumptions around that it sort of ties in maybe some of the folk quantum that came up in some of our previous sessions about what people think that is so i i'm really excited to see this folk idea being extended because i think that could be really useful so thanks thanks stephen and to dave yeah yeah dave douglas i'm retired from programming machine translation software um now i get to get really busy working on a history of cybernetic learning theory project for a professor at carnegie mellon and all this good stuff here thanks steve so matt thanks again for joining we'd be really happy to hear some context and we'll just take it from there and anyone can raise their hand or type in the live chat if they have a question fantastic thanks so much for having me uh so uh the context in this paper i suppose it's like i wrote this paper uh and during my phd over at the university of edinburgh i was supervised by dave ward who's an auto product and activist uh andy clark uh who i'm sure all of you know and uh julian kaverstein who's up over at the university of amsterdam and the the focus of this paper uh was pretty much my obsession on the notion of collective intelligence and what collective intelligence might be and how cognition uh has developed say the evolution of cognition with with a background of um or taking into taking into account the fact that cognition is uh in its most basic form something that's going to be collective in the first place uh so it does pull at there is this kind of immediate um tension with how we look at cognizers and how we look at cognition uh in the kind of folk field with typical uh say typical cognitive scientific kind of research i suppose a traditional should i say uh and the point of this paper was to okay but look at uh the kind of evolution of cognition if you if you will through a particular lens using some of the kind of the tools conceptual tools that were provided by our seth morty and and uh john men and smith and see exactly how this can be uh formalized and looked at through the perspectives of the free nature principle with the process theory active inference so that that's that's pretty much it very concise summary major transitions which is the zothemary and maynard smith line big qualitative changes over evolutionary history and then just like you kind of brought to at the end how do we land that plane with maybe quantitative differences or structural differences in a micro process theory maybe active inference under the free energy principle um so shannon or dave or steven feel

free to raise your hand um otherwise maybe a first question that came to mind for me was like how do we research or understand these folk beliefs scientists are folk too and is this like sort of the mythical average citizen what are these beliefs and how do they come to be that way why are they the way they are these folk beliefs about biology because that sets up a lot of the discussion and what's relevant of the contributions of the paper absolutely that's a very very good question very good question book beliefs we have kind of obviously kind of folks psychological beliefs that take the form of actual uh for psychological notions that take the form of say beliefs and desires and these things i think uh give us a good a good very good manner to in which to actually approach the notion folks thought biology also with beliefs and desires being particular type of constructs which play an explanatory role a psychological explanatory role these are successful in for example explaining or predicting behavior to us on a certain level and i think uh on a kind of first blush level so to speak and i think folks psychology and folk biology are kind of bed folk in the sense that biology is going to be based upon what we initially see via like in our everyday experience of the world for example is what we initially so we're going to base the notion of an individual for example on the on on the ideas of things that are going to be easily individuated uh and like biological systems that are going to be specifically reproducers and this is where you come into the the notion of kind of the kind of darwinian individual as it was called by peter godfrey smith uh where individuals are going to be demarcated by their ability to be categorized under under three different types of that combined or joint three different three different kind of categories which are going to be they have to involve variation heritability and some type of differential fitness which is just natural selection right and uh and this type of individual has been kind of uh saying we've invested into this notion of individual and by the biological sciences up into a certain point up into a certain point because it helps us answer helps us approach certain questions and make certain i suppose make certain inferences about about systems over the long run but it falls short and i think this is again one of the grounding points of the paper it falls short and explaining certain types of associations and these associations namely are going to be symbiotic associations and this is a quite important uh i say transition as as as we spoke about before uh for the simple fact that without symbiotic associations we don't really have a manner i think i think of explaining multicellularity right the the transition from single cellularity to multicellularity now that this we've had various theories and i think one of the theories that kind of one of theorists that started this was lynn margulis over at the university of madison wisconsin and her research was fundamental and and when she suggested that for

example you have two two prokaryotes one envelope the other through some type of phagocytosis and they create some type of relationship over time there's just thought as ludicrous and it did really push against this notion of darwin darwinian individual uh however over time we've come to accept something like this uh and uh i think this is like the idea that you have some type of bacteria that was enveloped by a an archaea in order to explain in order to actually think about what we have in terms of this particular type of unit uh we really do need the notion of uh moving beyond moving past our winning individuals and two ways of doing that obviously i think that the paper tries to point out this uh and it's these are not the only two ways but one way is one way of doing that is through kind of the metabolic individual which is something that peter gabriel smith himself kind of um uh tips his hat off to uh which is just to say that metabolic individuals are individuals which they can be delineated by their metabolic machinery and having uh kind of autonomous metabolic machinery and other other individuals can play a part of that machinery and thus you have uh two systems that may kind of be metabolic coupled to one another making one system and that's one way to go about it the other way to go about it is i was introduced by thomas and this is a kind of the immunological version of an individual or a physiological now again these are the only two types of individuals or two ways to delineate individuals outside of the kind of more traditional folk biological notion of individual but maybe that's something that we can talk about perhaps there are other ways that it might be interesting great points and so interesting how the map replaced the territory the map of the sort of darwinian triad defining an individual in the darwinian sense and then people project that back out and say reality is explained by this partitioning scheme yes so very cool steven uh with the raised hand then anyone else in here or in the live chat go for it stephen thank you yeah i'll be interested uh i think this is really good points about these different ways once you start to have um these interactions and these repres reciproc reciprocities across p you know then pluralism starts to come into the picture yes and i i'm wondering how you see um this moving between fields of practice because i wonder if like people in the biological sciences and maybe those who work with agents would really be getting this right and then people who have a theory about thinking they know what biology is maybe physicists who maybe think they know what everything is um then have you know they have a they have a biology understanding of biology just like maybe other people have a physics understanding of quantum physics so i'd be interested in that and this might even bring up some work of chris fields that we had on here as well in terms of other ways of approaching this multicellularity but i'd be interested in

your thoughts first about you know that tension even within the sciences that's a really very good very good point i do think this is a uh before i hit that point directly i do think this is a call for and just interdisciplinary work it's an absolute call for interdisciplinary work this is something uh that's that's it's i think trying to actually use psychological notions of biological notions or physical physics notions uh we will turn we will have some success doing these things however once we run up and there are going to be certain constraints uh with doing that and we'll run up to problems run into problems and these problems i think can only be can only be avoided or actually uh addressed when using various disciplines together and i think this is one of the things that kind of the free energy principle actually offers uh as a kind of it's an opportunity to work interdisciplinarily on on various uh various phenomena various processes uh and you're absolutely right in saying that like okay well uh how how can for example in biology you might have a typical assumption about what cognition is so we'll have a lot of biologists for example there's a there's a there's a type of i that we were speaking about before the before this there was a a notion of basal cognition that has been kind of pushed uh within the last 15 years and a lot of various kind of research programs scientific research programs whether it's going to be plant biology or microbiology has has kind of fed into this which is it's a lovely program in the sense that it's questioning uh how we actually look at cognition and these biologists that are kind of taking these cognitive terms and using these cognitive terms one of the things that you know uh i think that pushes against this is going to be kind of a traditional psychology that uh refuses to uh with lack of lack of better words i suppose which uses to actually engage in some type of interdisciplinary discussion so it's it's a happiest psychology that's happy using for example um more focus focus traditional and folk uh psychological notions uh with these kind of limitations i think we're going to actually be stuck we're really going to be stuck um and so i i guess to summarize the point is is that it is a call for interdisciplinarity it is a call for kind of questioning uh or putting aside the fact that we can give uh satisfactory explanations but whether or not our satisfactory explanations are actually the explanations or if there is just one explanation uh it that's that's the question that's the question and i think that the whole notion of uh the whole the whole idea about uh biological individuality is that you can be a once you take into account various various angles from various disciplines it's fine to be a pluralist thanks i i was thinking of just a pluralism of what the word folk even means some people might think oh folk chemistry that's being wrong that's the middle school level of understanding chemistry and then the experts they know about chemistry that's folk versus expert but also it seems like focus being used in a way

appeal to first principles like appealing to directly intuitive observable processes and um what is active inference if not a first principles approach there's a way to bring in some of the technical expertise the computational backbone but also fully retain the conversational and the casual approaches that appeal and spread to people in their understandings of different yes so dave with a raised hand and then anyone else with a then yes go for it dave yeah there has been so much good work done by people outside their specialties i guess erwin schrodinger may be the most well known of uh in that but um ben malka the swedish uh neurophysiologist uh wrote a lovely little paper from about 1999 i don't think it was ever published about how the development of the mobile eyes in one extremity of the early vertebrates has forced neurological developments that have resulted in consciousness simply because having mobile eyes creates such an enormous burden on sensation that you can't just use a simple nervous system otherwise it's just overwhelming have you looked at any of mercury america's uh work no okay i'm gonna have to send you that please do yeah yeah and um also mark solems uh who's a direct collaborator with uh professor fristen just wrote a popular account of um active inference from the viewpoint of his own work with america and yak poncep and antonio dimazio on the effective uh foundations of consciousness and especially his own thesis that in freudian terms it is the id that's conscious and the ego the cortical and the neocortical processes only gain to become conscious when it really has to because it's so expensive metabolically and in terms of distraction and he's one of the few people who's actually aggressive enough in trying to go out and market his dissidence from the received wisdom and he's also he's also did a uh an amazing update of the original project for scientific psychology in terms of fristonian active inference so first word the last word of that 1895 freud essay is in 2017 terminology but still in exactly freud's order of presentation wow yes please do send me that paper i would love to read it awesome so stephen with a raised hand then shannon or blue welcome and anyone else in the live chat of course yeah just uh also this this idea of the folk psychology and i agree with your point about interdisciplinarity and it's i think it's really interesting is how the experts perceptions of the world then gets translated into a folk publication which then becomes the kind of landscape which pretty much everyone then starts to incorporate and one one that i found really interesting with active inferences i was looking back at pre uh older ideas of vision before they discovered the retina and the image on the retina and they actually thought of the eye as projecting out and sort of feeling the world if you go back like 500 years and then we were told by the experts how it is with the retina and the image projected like a camera so then folk psychology moved to that or folk perceptions move

to that because the experts said that but actually it turns out that the original folk psychology or the original perception is actually more accurate now that we're looking at circades and uh all of that so this is an interesting what is folk and what's expert and where does it come from absolutely it's a both and because it's not like the retina emits the photons that it then senses back it's not doing echo location but there is a tale of two densities there's a generative model and a recognition model and there's a process that's a conversation which we're going to get to with users and resources distinction so it's like there is that bi-directional conversation and then if this direction is folk and then that direction is expert in certain social contexts which direction is going to be prioritized as an explanation and the other direction is going to be seen as a counter current literally going against the stream mainstream how does pluralism allow us to recognize both directions in that case yes stephen yeah that's a good point daniel um that that i do a bi-directional and one question i i wonder sometimes i often work i'm i'm sort of connected to people who are i'm seen in the world of coaching and psychology as well in a ways and the arts where these things like bayesian brains sort of did come through there about 10 years ago but in a very but it's it's normally come from research which has been interpreted into a book by someone from that field which then gets pic which is still quite technical which interpreted by someone into something else and by the time it's rolled down into some other areas these notions can be quite quite watered down you know um so yeah this this question about how to you know have that conversation when some of the stuff is quite cutting edge from even people in the the fields themselves um and how to get it used in the real world is is a big one absolutely very true so dave with a raised hand and then anyone else i think it's in the panda's thumb where there's a rather exasperated essay about how you'll have an innovator and if he's really lucky he'll have a few people that work with him that fill in the gap but typically and typically they don't he doesn't even have one level of competence it just immediately goes into the full press and becomes just a complete mush with all the almost all the original valid insights just buried under full foolishness possible so i guess just we paused on slide because this concept of individuality is a central one here and we mentioned a few of these different ways of looking at individuality and kind of contrasted pluralism the idea that multiple perspectives can co-exist and that that's okay versus i guess absolutism or modernism that one of these or some other unlisted criterion of individuality is like the the broader category or the right one um what is on the table with definitions of individuality like why would it matter in a technical capacity or for just all people what what is being discussed or what matters depending on how we approach this question of individuality to matt first and

then to right i think there are a few things that we can say why it matters one thing is that we have before we engage in some type of investigation about a particular type of phenomenal process it's quite good to know how to individuate what those process what's what are involved in those processes or what that process belongs to the relations between elements in a network pretty much right in order to understand those elements or to locate those elements we have to have some type of criterion in order to in order to actually even think about uh the kind of larger structure of a network and the relationships between them and i think this is biological individuals uh they they actually play this role and i'm reminded by something that uh gregory bateson said right that in order to actually understand what the mind is we have to actually understand how to delineate things with minds but those delineations are going to be completely dependent upon uh the the processes that we're studying those delineations are going to actually involve the processes that we're like so it's going to be a process dependent or a phenomenal dependent individuation and that doesn't say that doesn't mean these kind of individuals are going to actually uh they might contrast or conflict with one another in any type of logical matter but it's just putting in place as a type of pragmatism in terms of uh where we delineate things and i think biological individuals provide us with a manner of talking about for example specific phenomena now if we're talking about replication we're talking about for example competition uh with respect to something like replication then we might want to actually focus on darwinian individuals however if we're talking about the evolution of cognition or we're talking about the evolution of multicellularity or these major transitions in evolution we might want to think about individuals in a different way so that's that's that's what i think thank you matt blue and then shannon so hi matt it's nice to meet you i love this paper found it super um inspiring and um you know i think that the concept of an individual um is closely related to maybe the concept of autonomy and also the concept of emergence and so as and especially i mean it's also of course related to the concept of a cognizer but but cognizer aside so i've heard a lot of definitions thrown around and an individual um i liked in the paper how you described emergence with the circular causation because that like echoes a lot of um i mean it's the the information theory of internet individuality paper by um krakauer and that group and it just echoes a lot of i mean what's required for an individual so i i just i'm thinking about um autonomy and and it's one of the things um i mean it depends how you define it and in like the artificial intelligence world there's no such thing as an autonomous system i mean i mean really we program in whether we want it to learn whatever it wants to do but i mean they just don't little baby robots aren't just set free with no goal there's always a

self-directed or a human directed goal kind of programmed into this system or maybe i'm wrong maybe there's examples where it's we just create robots for for fun or or artificial intelligence just to create it but like an artificial intelligence system doesn't really have that same autonomy or at least i don't think about it and so i just i'm kind of curious as to you know in the paper you kind of hint it out or i don't know if you said explicitly i was kind of flipping through looking for a quote but you know you hinted at like maybe a cell doesn't also doesn't have that same kind of autonomy like so i i um i i want to challenge that in in terms of active inference and the fvp like i think about it always as a skill-free architecture and a continuum why is a cell not autonomous but an organism is or maybe an organ is is an organ autonomous or where do you draw the line and why are you making a line in the sand around a system like that fantastic question and uh i agree with you i agree with you that's one of the things in the paper that i think that i pushed i did draw a line i didn't try to draw a line that was too deep in the sand uh and i think in one of the footnotes i do say something like okay well i'm not suggesting that a mitochondria is going to be something that's non-autonomous however autonomy is something that also varies in degree that i think a certain level there's a certain level of autonomy uh that it's going to be providers that provide a a different degree of of i say degrees of freedom with which some system can interact with its environment in ways uh and that integra that environment provides perturbations uh enough perturbations enough kind of enough surprise such that that that whatever that system is has various strategies uh it has to have various strategies behavioral strategies or we can talk about this in terms of active inference uh and in order to actually adapt to that particular environment now i'm not this this gets us into this notion of adaptive active inference versus uh kind of like the non-adaptive active inference and i think the way that michael and julian have thought about if i'm not mistaken i have thought about uh adaptive active environments is that it is based on a certain level a certain degree of autonomy and with respect to cells you know i agree i do agree with you cells do have a certain level of autonomy however cells are again dependent upon other cells uh in a way that other systems other more autonomous systems aren't dependent now you'll say of course we're dependent upon our environments we have to have oxygen there are certain constraints in our environments that we have to um we really have to kind of like our environments have to be a certain way so we are dependent upon our environments but it's it's the distinction i think between being dependent upon other systems uh and cells for example bodily cells are dependent upon other cells uh not only uh say that plasma but other cells in order to actually maintain themselves but i think there's a difference in degree so that's that's

that's and i shouldn't again i should impress upon that this idea that i'm not saying that cells are not uh autonomous but most certainly i think there's a there's a difference in degree and that difference in degree does uh does matter uh cells might eat cells might even be cognitive to a certain degree and this is something that some someone like mike 11 michael levin suggests and i and i agree with him but i suppose the notion about adaptivity uh there's a degree of addictivity that comes along with us uh the emergence of a certain kind of uh degree amount of degrees of freedom so cells are dependent on other cells but aren't humans dependent on other humans i mean we're born as an infant and completely reliant upon our parents to provide everything for us for many many many years and and even as adults i mean still we don't exist in a bubble i mean perhaps we can go off to a cave and exist in isolation but but i think humans are dependent on other humans as well so maybe it's a degree of dependence or not or i mean i i really feel like there's a continuum here even in terms of adaptive active inference and mere active inference i mean and i fall in with mike levin in in a lot of the ways that i think so yeah yeah i i think i think that's right i do think that's right i think there's a distinction to be made also though between kind of uh developmental dependence uh and homeostatic dependence right and and i think homeostatic developmental dependence can have an effect on homeostasis homeostasis most certainly but those are two things that might not actually uh always always overlap and uh for example sure we as humans absolutely we have a long developmental period in which we're absolutely dependent upon our parents or being reared by some other human or some other animal absolutely and that happens that when we actually uh when we when we get older we probably are dependent also i absolutely agree with that but there isn't any kind of development with a cell that can be mapped on to uh dependence or non-dependence it's always dependent thanks matt shannon that actually uh flows really well into my question blue that um i know your paper you specifically say we might want to talk about humans or groups of humans but we're not going to in this paper um but i'm going to ask about humans and so when you're talking about um squid and it's bioluminescence um and they form a temporary unit in which they are together an individual um and then it sheds the the the bioluminescent cells during the daytime and then they breed again afterwards i'm wondering if there is a time like a critical time period at which units would have to be uh have this symbiotic relationship with each other or if there's a necessary strength of the coupling or a repetition of how many times um the organisms need to meet up and if that would change for for this this animal compared to systems of humans because there's a certain way in which um the the humans that you interact with in everyday life or if

you're co-living with someone even your metabolic states become co-regulated by your sleeping patterns with each other or stress and hormones between different humans are affecting each other when you're together at a fairly regular rate and so i wonder if we might define individuality or collective slightly differently depending on the system we're looking at and the type of coupling that's occurring i think that's a great question chad it's absolutely i would i would say yes i think this has to be some um obviously that you know about the the empirical uh literature in terms of kind of entrainment and i think this is like that's a really good point to make in terms of having those types of interactions with humans whether or not for the for the bobtail and uh vibrio uh there's a chance that um like out after the vibra actually colonized the light organ they're typically not taken in again uh that once they're expelled they're out and they might be taken in by another squid or juvenile squid but they're but the vibrio that continued to colonize are the ones that are left over it might be the case that the ones that uh continue to colonize for that shorter period of time might become the relationship with the squid might change to a certain degree but i don't know any uh any empirical evidence and i'm ignorant about this point that suggests that's the case but i think that's a really good really good question to ask a very good question to ask so blue and then i'll make a question i think dave was first oh yeah go for it dave at them yeah they're from dr alan shore has done some um i think controversial but really evocative work on the formation of the not just the personality but actually the neural functioning of the newborn in the dyadic relation the intimate dialogue with the mother face play and the rhythm of moving in and moving back and regulating each of them regulating each other's expression of interest and each other's degree of involvement and at a really different scale gerald edelman's topo biology talks about the cell and the tissue as being just so dependent he points out look we have almost no chromosomes we have almost no genes compared to how complex we are where does the regulation come from it comes from adjacent cells adjacent tissues right thanks steve blue so um shannon kind of what she was saying about um becoming co-regulated say if you live in an environment with someone like it's a super nice transition to something that i have really been wanting to discuss uh in the context of this paper which is like okay you become co-regulated with someone that you're roommates with and and co-regulated perhaps by the environment but also perhaps by the microbes that live in our environment right so there's like the entire micro microbiome concept that is is fundamental and key to humans and i don't know i mean i'm not like a a huge studier of symbiosis but i'm not sure if the microbiome like i know we depend on a healthy microbiome i mean for healthy like brain function and you

know just regulation of our own internal like uh metabolism and so forth is that a symbiosis or not um also like the concept of the holobionts which is like introduced by fields 11 if you've read that paper they're the scale free biology paper i mean it is like to what and that they they talk about um physiological individuality there as well so is that symbiosis is it not is symbiosis it's also a spectrum or is it like a you know a discrete category of things and i also think about like symbiogenic viruses which like you know the very early viruses that it's like you know effectively like instantiating speciation right like the viruses insert into the genome and you know developing species and so forth and and it's beneficial to the virus and the host right so so anyway i just wonder like what are your thoughts on that or or or um if you have any insight there absolutely that's again a wonderful question boo um with respect to kind of symbiosis i think there's some there's been some work on that and some very i think uh convincing work done by queller and strassman who suggests a particular type of xy locating symbiosis on uh kind of an axis pretty much so you have uh cooperation i believe on one part of the axis i have to have to look it up again actually i should look that up um but it's uh you have varying degrees of symbiosis uh and again symbiosis in the in the particular paper that we're discussing now i've concentrated on mutualisms but you have various forms of symbiosis uh that i think you've actually pointed out previously uh that can be kind of modeled quite well with kind of game theoretic uh and this is going to be is going to rain from parasitism to uh communalism to kind of um kind of mutualisms and i think like these are all again degrees and these degrees can change uh there have been quite a many theories i suppose and i think squadron strassman have also suggested this that uh depending upon the condition of a system uh with respect to another system say their symbiotes a parasite can turn into a mutual a parasitic parasitic relationship can turn into a mutualism and vice versa so these things move these types of things move uh and if you think about this is an automatic thing that we can even apply this kind of cellular biology when you think about something like the the origin of cancer when you have cancerous cells they're they've actually grown within a host right they're part of a body uh but end up being some type of competitor and i think this is quite interesting to think about this it's like through the lens of symbiosis obviously it's it's not that's that's not a symbiosis but we can use symbiosis to kind of think about but with respect to kind of uh thinking about host cells and parasites our whole cells and mutualisms uh there's like there's there's many many different kinds of ways areas of the spectrum that one can actually analyze uh our relationship with our own cells and our relationships with uh with micro microbiota microbiota like scott gilbert obviously you know and scott's work is is quite

incredible he's the one i believe who introduced the notion of the hollow biome the hubble genome excuse me in halloween i think this this notion about uh hollow buoyant is specifically it includes symbiosis but includes actually a kind of community a particular kind of community of various systems uh that themselves qualify as one system can qualify as one system or having you know if you would put it in a different way they they create a stable state right a steady state themselves and uh one that's kind of resistant to perturbation that's that's the matter in which i think it's like it's it's a very easy matter in which i can think of uh blind as as an individual uh whether or not one wants to call two two symbiotic say symbiotic uh systems that are in a mutualistic relationship a hollow bond that's another question but i think typically it's been applied to um more than two systems thank you matt so for shannon's questions about you know the co-regulation and humans and trust it made me think about how one-off games lead to kind of community or population-like game theory whereas repeated games over evolutionary time like in the case of the squid and the vibrio or in the case of the eu social insects that leads to organismal-like dynamics and then what you just described and what um levin also got at in the paper is the breakdown of for example organismality into population of cell versus population of cell dynamics so it's absolutely um a cool framework and then also a really important part on figure four is you have the peak of integration and then greeting serval the peak of integration um and then that's the y-axis so it's like instead of saying we're gonna say which organisms are tall or short out there in the world it's like you do the population level survey and they range from you know two to four feet and you say okay it's in the upper 50th percentile it's tall like we're renormalizing our comparison stick to the maximum integration of the squid bacteria so we're not comparing it to some other system we're re-normalizing and we're able to have a system-specific understanding here it's relatively more integrated here it's relatively less it's a second question whether the peak of integration of the squid and the vibrio is like the integration of some computer program with another computer program maybe it's apples and oranges maybe we'll have the right framework that allows us to transmute apples and oranges but here you can just relativize to the system itself and that's like very specific and powerful so stephen with a raised hand and then anyone else yeah i'll give you a chance to get a breath because we've thrown a lot at you but um the you mentioned allostatic and homeostatic forces if i imagine there's the the so the so you've got that difference there between something trying to maintain or trying to try to move forward and change and i'm interested in in how that reflects um so for instance i suppose you could have within a set of cells you could have some cells or some organs you know trying to

maintain homeostasis and others say even maybe a parasite is trying to anaesthetically enter that world and create a niche for itself um and i think that that that question about that that balance between the two is a useful divide or a useful way to think about the same cell and of course in cells they change to become different types of cells so that's something that whole organisms have less ability to be completely completely different types of organisms you know they can't um you are what you are i suppose that's one defining thing about but you can still change your focus from homeostasis to allostasis i think that is quite common and i wonder what your thoughts are around that um and how that that itself could be a useful way to frame things that's really interesting uh the the man in which i'm looking at kind of a homeostasis uh and uh kind of coupling of homeostatic regulatory systems to allocate regulatory systems is based upon kind of a more traditional idea that homeostasis might be considered more reaction or reactive where allostasis is anticipatory right and then allostasis takes a particular form it's quite interesting with respect to in the context of active inference because we talk about expected free energy minimization and this is something that uh one can tie together with the notion valley stasis where uh and when you start to think about kind of homeostatically coupled systems uh with respect to kind of symbiotic systems might be playing a kind of a larger rule uh and being able to make further and further of being able to make inferences further and further into the future and i think this is something that you've you've suggested you've looked at previously actually on a slide uh mike levin's cone slide that blue and daniel actually had up previously and i think this is a quite interesting manner of thinking about uh kind of a transition perhaps we've always been allostatic systems living systems have always been had certain of certain uh capacity to anticipate however when we start putting systems we have a larger capacity or more capacity to anticipate further and further in the future and i think one of the dimensions on this particular cone is this dimension of alice like a kind of outstanding control uh this is this is my interpretation of michael's work uh and i think this is that's very important and again you'll see that there's that balancing that if one is given a type of approximate bayesian inference that's going to be uh you know what we suggest is going on uh you have to actually balance this with priors which are going to be respective to the past so you have another type uh another axis on the cone and as these things expand sometimes the past is going to uh be a bit deeper on the cone than than the than the future but out of stasis is on on the top there and i think that like with with multicellularity with uh with kind of uh the interactions more interactions perhaps you will see a capacity a larger capacity to go upwards and in the kind of future oriented direction and this is this is

something again uh and i i appreciate blues bringing this out this is this is uh something that's again a matter of degree it's not a it's not a line that's cut it's a matter of degree just simply we know that there's certain types of uh microorganisms that that do that do like right uh and uh this is kind of outward anticipatory behavior whether it's i think it's going to be uh phaser and polycephalum which is uh the plasmodium slime mold uh various experiments in 2011 uh had suggested that you know uh fuzzerum can actually various periodic uh bouts of dryness this this type of thing this is an outward periodic behavior an outward anticipatory behavior that's not to say that anticipation is not happening occurring prior to that in the internal dynamics of systems that's just an uh kind of an outward manner of kind of measuring it again i i do think that this like it's it's a quite important notion allostasis it's perhaps all systems again i i would be safe at least my position now is that all systems are pretty much allostatic systems uh that kind of enslave homeostatic regulatory systems uh to maneuver invest their metabolic machinery and metabolic capacities uh to a particular type of uh state that perhaps they haven't reached yet considering uh you know a manner of of trying to kind of avoid perturbation thank you matt very similar to mere adaptive inference being more homeostatic more like the ball just in the bottom of the bowl and then increasingly anticipatory rosen's ecology adaptive active inference stephen go for it yeah this this ties in actually i'm a lot of work around um coaching funny enough the world of coaching and um team coaching is like and team coaching so and i think this also brings in you know these other types of clusters groups symbiotic um relationships beyond maybe what we might just think of as a system you know so the idea of a team um in a as a in a systemic context um is really interesting because they and this might be how allostasis in a social context starts to enable that larger horizon so each even allostasis is sort of stasis and then working together you may start to break your own need for stasis you know and it becomes some other organization and i think it'd be really interesting and looking at how the way that team coaching either for performance because there's performance orientated team coaching or there's developmental orientated team coaching but that idea of and then the enabling context becomes what we would call the ecological niche i suppose becomes as much the focus of attention as the system in that makes if that makes sense and i think this this might tie into what you're saying thanks steven so dave i know that you had a little bit to share but i'll just bring up one point related to this definitions of individuality and then several anything you want to ask feel free to raise your hand we hear a lot of terms like autonomous and self-organizing which sort of highlight the um ability to cleanly partition a system like people will

say the embryo it's a classic self-organizing system right except for the context of the womb and so there's sort of on one hand the desire to individuate and partition and look at agentic systems agent-based modeling autonomous and self-organizing and then that is critiqued by the ecological perspective or the relational perspective so there's sort of a dialectic between like it's all individuals all the way down and it's all relationships all the way down we see that debate in particle physics we understand it less than we might some of the biological systems are in a different way but it's um like two pillars that we're navigating between and it is a continuum in between so that's why it's such a nice discussion so dave if you're around um feel free to um any time you're ready you can go for the conversation theory otherwise blue go for it okay sure um now slide yeah slides 18-19 we inserted that into the slide show that we prepared for dr sims and these slides bring in some matters that he doesn't address in his paper so to justify that but not to paraphrase the insertions the free energy principle and active inference jointly constitute a relatively mature macro level framework for describing living psychological and other complex self-stabilizing systems as of the moment no comparably mature micro level framework has been both proposed for and coordinated with this macro level of analysis i suggest that the basis for such a needed micro level framework was proposed by my teacher the late cybernetics professor gordon pasque of brunel university in the form of conversation theory matt's essay how to count biological minds considers the puzzle of whether an individual can come into existence and pass out of existence repeatedly that's account of this puzzle and please excuse me if i stray from over simplification into parity his solution in terms of degrees of integration is a perfectly acceptable insightful at a macro or molar level the metaphors of quantity and elsewhere probability nicely express falsifiable common sense attributes of real-world but if one states this paradox the paradox of emergent but transient individuals in terms of conversation theory then one speaks of a conversation recurring over time between the same conversants this terminology makes the proposal more intuitive less shocking and more readily assimilable for readers better yet familiar ideas attention rhythm of discourse punctuation of a conversation through understandings or through agreements to disagree they all play their roles now what place would be occupied by the special disciplines that have enjoyed detailed treatment by conversation theory as a micro theory during his career professor pask carried out an array of experiments studying human learning of complex subject matters configurations using configurations of software and hardware to administer and collect data from those experiments and develop a vocabulary allowing elegant description and restatement of the results of those

experiments moreover he developed a theory of micro level learning teaching as a single literally irreducible complex of activities regardless of whether they were realized by living organisms by machinery or by complexes of those two fabrics i see many deep and broad analogies between conversation theory and active each addresses important aspects of real world phenomena that assist in a systematic way such that each treats of matters that are not so well handled by the other i believe that active inference and conversation theory can enrich and streamline each other by carving natural science scholarship closer to its joints very nice thank you dave so maybe matt any thoughts on that and then we'll go to blue i have to say i have to say that i'm uh the the cybernetic theory that i've concentrated upon have been most influenced by uh it's it's very limited and conversational theory it's i'm absolutely interested in seeing more of this more of this i suppose analysis and understanding more of this analysis so i'm quite ignorant about conversation theory and it's like he most certainly piqued my interest yeah let me let me think about this for a few minutes i'm still kind of internalizing what you've just said yep and one note on the conversation a lot of times cybernetics falls into that individualism mode like it's the one self-guided missile it's not the conversation between the flock of birds so cybernetics um isn't just about individuals it can be about collectives and it was our colleague alex who told us um via one of his teachers levinchuk just yesterday that um cybernetics died when it could explain homeostasis but it couldn't go beyond so we're talking about the transition from individuals to collectives and from mere homeostasis to adaptive allostasis over developmental and evolutionary time scales and so it's sort of like you can have homeostatic individuals and then they can have emergence towards allostatic collectives and then what is that process of emergence it's a conversation or potentially something like that using our folk understanding of interacting pieces which are us so um my question is maybe technical but also uh maybe serval has some insight to provide as well um so i wanted to ask here in the beginning of your paper you talk about free energy being an additive quantity the free energy that arises at the macro scale is only the free energy that there is for the ensemble and may be quantified by summing the free energy associated with each constituent micro scale markov blanketed system so is that true and and like does that account for emergence because i feel like there has to be some like greater free energy minimization and so like i i'm not i'm not well versed in the technical like mathematics of the fep um because i think about you know like just in terms of the bioluminescence of vibrio fisheries right right there's like a criticality that happens like almost like a phase transition that the system undergoes to become bioluminescent like after there's a certain concentration of

bacteria in the light organ and so like does this criticality this phase transition does this not like to minimize free energy to a greater degree like or is it really just the sum of its parts wonderful question really wonderful question i do think that okay there's a might be a distinction in terms here in a distinction in terms of like what uh the free energy uh of the actual constituents as a group have as an ensemble have and a distinction between that for example and bioluminescence as a particular type of free energy minimizing strategy it's a i think it's a fantastic question i'm getting the the the notion of kind you know the summation of free energy from uh palacios and aronstead this i think that the pretty much the idea is that i'm not sure if there's going to be any well i'd like that maybe i can ask you blue do you use do you see that there is some type of inconsistency between saying that for example free energy is going to be an additive quantity and the fact that there can be some type of emergent property that is not going to be reducible to say the the constituent relations and and the elements do you think that there's some type of uh inconsistency there well so i definitely do think that free energy across scales is additive like i i agree with that wholeheartedly but but i don't think that it reduces to the mere sum of the parts and i mean this also um it's the the work the recent work of rosas and also eric hole looking at the partial information decomposition over the um five metric that's used in integrated information theory of consciousness and so this kind of speaks to that there there's it has to be some mathematical i mean i just feel like i don't know that there has to be but i feel like there should be some mathematical quantification of more so it should be a greater degree of free energy reduction when the system is more than a sum of its parts and so i think that that's an argument that you make in the paper as well that it can't be it's it's not reducible to just the component parts so is the free energy then just reducible to the component parts here because that to me seems inconsistent okay okay the matter would i think uh i i see absolutely see your point no matter what i'm thinking about this though i said okay well there's a very informative paper by santos he's a philosopher he he actually deals with immersions uh and it's a really nice way i suppose i really like this uh the paper that you suggested about kind of using the five metric uh and various manners in which to like to do like decom information decomposition over various time steps uh in order to think about some way to some some metric for emergence i think there's been various metrics for emergence i haven't picked my favorite metric that's something that i'm still getting into but i think this is it's quite interesting to think about okay well uh what makes an emergent system is an emergency system going to be more than it's the kind of relationships and elements of a network and what makes it so what makes it so it's going to be the relationships themselves or

or is it going to be something uh the combination of the relationships or is it going to be the interaction between the system the emergent system and the environment and i think when you start thinking about the interactions between the emergency system and the environment there's there's a distinction between what free energy is going to be minimized uh or and the free energy of the system itself does that make sense is it there's a distinction being made uh and perhaps i'm wrong about this but emergence is not something and one of the reasons why you have an emergency system it's going to be for the simple fact okay well you have it's not only the interactions of the parts and the parts or the elements it's going to be it's going to be the interaction of a particular type of persistent and i think this is the this is the idea you have a persistence uh with a particular environment or with an environment and it's the perturbations within that environment uh that occur along with that system in which kind of free energy is going to be a free energy that might have to be minimized is going to arise and i think it's like so you might say of course that the the the system itself uh when it has component parts and relationships between those component parts is going to be uh that system can be thought about in terms of an additive or summation of the free energy of the various parts and relations in that system but that's distinct from that system's engagement or interactions with this environment then you'll have extra free energy you have an additional amount of free energy and maybe that's what that's that's that's another way a matter of thinking about okay well uh and an as measured i don't know if you can measure emergence in terms of free energy but of course we're going to be able to measure it in terms of some type of mutual information very nice thank you that's i think that was a question there i don't know if you can energy like you said and you might have to look at something like mutual information and i was thinking that perhaps the free energy that's minimized by each individual system then when they become a symbiotic unit or something might end up being identical and again i'm thinking of humans again but if humans are walking and they happen to start walking in step um that's gonna look like an identical oscillation as as they're coupling with each other so there's might not be more free energy that's minimized by this coupled system walking in step together um but it might look like this the same free energy once they're a coupled and one thing that we talked about i think in the dot zero or around there is there's situations where the cooperation the coordination increases synchrony so just increases the mutual information of the humans walking in step but also you could imagine anti-phase relationships which is kind of like division of labor or task allocation where the emergence is not due to um copy and conform but the emergence is actually you know you go left i go right and so in both of those cases

there would be a change in the non-equilibrium steady state which is what is introduced throughout the paper and so that starts to build the road towards quantitative discussions and you could take a system that you suspect already has emergence in nature and remove a part that's kind of like the gene knockout approach so gains of functions and loss of functions and all the other ways that we study function can now have something like a common grounding potentially using these metrics here dave and then maybe matt if there's a response otherwise stephen thank you okay now blue asks is free energy additive there's an exercise that i'd suggest it should bring that question to a more down-to-earth and more poignant focus take all the discussions you've heard of free energy and substitute for the phrase free energy the word anxiety it already is that way dave what do you want me to do [Laughter] what would that show though or do you think it's a natural fit or could it could be could we put any word there what do you think it means well so just using anxiety like it totally leads to the mob mentality and like ultra paranoia and conspiracy theory so like definitely additive possibly multiplicative though like could be exponential [Music] yes steve yeah there actually have been uh clinicians that have really used free energy although they didn't use the term as probably before the usage was was coined william glasser in control theory and his earlier work uh really has this quantum of anxiety that these patients had to manage these tended to be out of control criminal adolescence so they were pretty screwed up um and he found ways to get them to rein themselves in and control themselves but think about the blitz effect london blitz you had objectively tremendous stress on everyone all the time but there was almost no hospitalization for shell shock among civilians including the air raid wardens that were in digging you know they a building beside them get bombed and they'd rush in and try to find people and drag them out extraordinarily low rates of psychosomatic disorder there's a certain there's multiple paths you know people people's brains are really complex there are certain ways that people actually do get shocked into rationality i've been in situations where the right thing had to be done right now with no hesitation no anxiety no blame just do the right thing and there was no question just do the right thing uh whether it's whether it puts me in danger or not you know a child's endangered you do the right thing so it's not predetermined uh but typically yes what does happen is uh a thing called reorganization it's a technical term that glasser uses it essentially means going crazy and in the really worst case the reorganizations get faster and faster and faster and more and more drastic as the personality becomes more and more simplified and that's something that kurt levin or lewin worked through in his work on driving children crazy by giving them more and more complicated tasks and under more and more

anxiety and he was able to make the kids personalities and thinking degenerate and simplify more and more and become more and more infertile and you're right blue that is one of the things that happens in a mob the it's terrifying to be in an even unified angry mob where everybody's mad at the same thing and they're probably not going to attack each other it's terrifying and people that keep going into that situation things happen remember um uh um the presentation two weeks ago on the mutual suppression of the limbic system the reticular activating system on the one hand and the prefrontal cortex on the other you get these things that look like beliefs they sound like beliefs they act like beliefs but they're not beliefs because reason has never been invoked the biological decision whether to incorporate this magician's spell into my feeling and my actions do you have to have reason to have a belief dave oh yeah in the sense that uh uh philip gerunds uses it yeah you have i mean that's that's kind of the way he defines it a belief is something where the prefrontal cortex has been involved it's been in the loop and there's way to bypass that um i asked him whether he's worked on the social psychology of this or the the ego psychologist he says no no i have enough trouble worrying about people with damaged brains and how that shows the bypassing of rationality in the formation of delusions but i think it works the other way um you may have seen my my shotgunning of the apathetic apothemic brawl by uh dr nathan rosenberg or rather gabriel nathan rosenberg he's an academic and i think kentucky he says look this is what's happening you've got this person with a a non-existent personality who's doing what the mob tells him to do through the mob's megaphone called fox news why do we like him so much because he's less than any of us he will always obey us so we'll give him power and then he'll give us what we feel like uh yeah and there's no reason of ultimate it's very much a bypass yeah rita labonz uh study and mackey's study on the madness of mobs and the mass production of delusions and crazes mccarthyism and all this unhealthy stuff thanks dave and that was the guest stream 10.1 with philip guerins and so just to highlight one point and then stephen it's we're really talking about control theory maybe the first pass the first term in the taylor series is about control i mean come on narrow the error and that's kind of definition one control as homeostasis even if there's a shifting set point but then again it's the ecological and the relational insight and just the emergence of interactions control takes on this broader anticipatory usage to control your body temperature is to sometimes do a lot of movements that from that first pass look like a non-sequitur like sprint so that you can get into the air-conditioned room that's the expected free energy through time so now we have a common grounding for talking about control when it looks homeostatic as well as control when it looks

allostatic or even innovative so it's very important to pull out some of these threads and also recognize their cybernetic origins and other areas too so steven and then anyone else yeah um these are all really good points i'm curious when we talk about free energy minimization i think it's it's interesting to talk about that from a systems perspective um however if we want to get back to the kind of organism so the key thing with free energy minimization that occurs at different scales you know physical scales number of units scale time scale you know teleological scales we think about these different scales but that reduction there has to be a way for that reduction to be meaningfully engaged through actions or interactions like the metric ultimately what's going to be filtering through probably in some noisy maybe backdoor way is that um if it's going to be sustained over time there needs to be some meaningful engagement over time so for instance you know the idea of the path that we walk starting to wear away the grass in our niche well it's because we walk on the path more than once that we have an ability to observe that you know if we only ever walk that path once it wouldn't have a way for any variation to be noticed but we notice the path being worn by agents over time so when we're talking about these different scales these variational dynamics in free energy it's not just a reduction because we don't get a number each time saying this was lower than last we just get some new state in the world and we have to infer if it's changed from some other state in the world and we it may go up it may go down so we may infer at times the free energies has has not been minimized and we might take but overall static processes for instance if we have a big room that's not tidy and we have a group of people and they want to organize it into a studio space you know they might work together initially putting in a lot of extra free energy time and and then eventually they know that they will be able to minimize their sort of systemic engagement as a group while it's not necessarily that they're each minimizing free energy as a system in the so this is almost where maybe challenging to get out of the biological sense and maybe come into some other disciplines where it's not all a system it might be more of a systemic context which is then more teleological so i'd be interested that that question okay where does this crop through this idea of being able to to have a pragmatic metric that links the non-equilibrium steady state of the organisms phenotype or individual so they're actually able to then infer something from whatever is happening to the free energy out there or in in there or around them and how that changes things if we if we frame it away from systems per se to something more systemic different uses of the highly poly semis word system systems of systems for some people are systems so that's why we um increasingly will try to be exploring the active inference ontology so that we know which pieces of the puzzle

we're talking about and then that will unpack so interestingly into different disciplinary contexts stephen so nice points um so in the last half hour i think there's a few different things let's maybe get to one of the key pieces and specific contributions of the paper which i would love to hear matt your thoughts on which is just first what's going on with users and resources and second what is the difference between uni-directional multi-scale integration and reciprocal multi-scale integration right um those two notions i suppose of unidirectional and multi-directional reciprocal and users and resources they're very much so connected the idea of a user we can think about this and i think stephen's uh to use stephen's example of an elephant path i believe that's what these things are called sometimes so you see these paths that are treaded and are typically you know used to refer to when we're thinking about kind of nice construction like past phones construction [Applause] this is something that is in the environment that's is being used say is a resource that's being used and it's the effect of previous action on the environment uh returning to the notion of so that that that would be a resource used by a particular cognitive agent or an agent that's engaging in adaptive active inference uh number one it has to be recognizable uh and it has to actually in order for the system actually to engage with that particular resource it has to be recognizable as such and that recognition can come through for example epistemic foraging uh or actually just uh be applying particular kind of a plan a model to a particular type of uh resource that has been applied previously um this is also the case with something like stitch merging right where you have uh something uh that's it's quite lovely where you have for example ants leaving pheromones within the environment uh and that pheromone acts as a particular type of mediator uh directing further behavior by agents of a similar type and that's again environmentally mediated communication or or what they might call an environment environmentally mediated uh conversation between between various elements of a system the pheromones themselves of course are resources in the environment that's going to be perhaps uh contrast with the idea that uh not all resources need themselves are themselves they have to be limited to being resources they can also be users also so uh to to make a strange case we can think about uh stitch mercy actually uh what if it wasn't actually the pheromone that was being used but the other act the presence of the other ant that was being used the interaction with the ant so here the two uh using using using perhaps and perhaps you have to tell me if i'm using this wrong that's terminology but you have two discussions right a cinder and receiver both are but both active senators and receivers and actually they're communicating something they're having a conversation uh but there isn't any mediation in between the two

uh that can be offloaded onto uh in this particular situation offloaded into the environment so the signal itself is going to be the direct kind of interaction between the uh rather than a pheromone now that in that case we can map that onto something that would be a kind of hypothetical case of reciprocal multi-integration right uh uh stigma rg just happens to be uh typically and in each cases in in the way that uh people degrassé uh suggested it when he was studying termites uh this is something that's actually involving a particular type of signal in the environment that's offloaded to the environment so it's going to be naturally something that's going to be a unidirectional multi-scale integration between a system of team very systems whereas for example when you have direct interaction between two cognizers or two systems uh two adaptive systems they can also of course use some type of resource there can be signaling and this is something that of course happens with vibro fisheries and the bobtail squid but there's also going to be some kind of other direct interaction between two systems and that's uh that's kind of the reciprocal awesome summary i believe the technical term would be converse ants for that kind of a conversation but it's a great point how um when you have an adaptive active inference agent and then a niche that's being modified stigmatically that does mere active inference like the spider's web that's unidirectional but still multi-scale whereas when you have adaptive and adaptive in discussion in conversation even again if only in a folk sense you have adaptive on either end and it can allow for a new non-equilibrium steady state to arise so not that there isn't a non-equilibrium steady state of spider plus web rather than them separate in fact there will be but it still is a different type it's like a double directional or reciprocal stigma g so just such a cool way to connect that blue and then stephen like again this like line between mirror active inference and adaptive active inference like is it like makes me want to attack it a little bit um i i think about uh the niche modification that for example humans do is extensive and transcends like generations and i mean work it like contributes a lot to the collective intelligence of the human species and so forth and i also think about um in terms of like the gaia hypothesis right like so if if the world is one giant organism and this is something that we talked about with mike levin when he came to the live stream so if the world is one giant organism is the niche constantly adapting to the species that are that are in it like is this a continuous adaptive active inference and just in in terms of transcending time scales you think about like these you know symbiosis i guess they're symbiosis but like relationships that develop between you know like orchids and their pollinators for example so i mean these are intense co-evolution things that happen over many many many times time scales and i mean we can think about the plant as a plant a cognizer i

mean that's debatable i would say to a degree like i'm not i'm not not binary here like yes no is not not really a good answer but but is the the niche may be constantly adapting to systems to a degree i think that's a great point really really great point um i do think the niche is adapting to systems absolutely but i would say that the niche is adapting in a different way and we can use of course we're going to have to be specifics because this whole adaptation takes a policy in a sense uh adaptation in in terms of what the niche is doing if if the niche is not itself okay perhaps you want to i can be accused of being biased interested in one sense if the niche itself isn't a biological system we can kind of perhaps push against this idea that it's adapting in the same way that a biological system would adapt to and may be one way of phrasing this uh i'm not sure if this is going to hopefully make a bit clearer my my point is that biological systems one way of understanding them is that they have set points right moving set points throughout a particular trajectory this particular trajectory might be a trajectory across face space throughout their life so you have a moving set and this moving set is something uh that's organized uh around a particular type of phase-based organization uh and it's recapitulated within a particular type of phenotype uh defines the phenotype it characterizes a phenotype or a genetic model uh with a particular say a a field of dirt we might be able to actually walk across a field of dirt or a set of cement stairs the uh it might actually over time for what a few hundred years you have these interesting stairs at the castle of edinburgh that actually dip it and these dips in this in the cement are evidence that people have walked there before but we tend to walk where other people have walked and this perhaps one might call that an adaptation to a certain degree but i think that might be pushing the limit on the word adaptation given that that set of stairs does not have a set point it doesn't have uh it doesn't have the same type of uh it's not the same type i suppose of of of self-organized system it's not a self-organizing system in that sense as such i would think that okay well uh maybe that maybe the niche does adapt to a certain degree uh but there's an adaptation with biological systems that are self-organizing and have organizational closure and i think this is quite this is quite important uh that is different than say um abiotic systems and this brings i suppose we i would love to talk about in a bit is that it's a question that blue brought up in the last um in the last kind of uh live stream the the manner in which we can actually understand systems or i identify particular types of individuals uh that are involving the environment so it's it's not actually saying that the environment is separate from the system otherwise we wouldn't actually be able to actually talk about the emergence of actually living systems we can get about we can get into that later though so i have to just push back on you um so think about

the earth and the atmosphere of the earth okay now this is not necessarily a biological system this is nitrogen carbon dioxide water clouds etc right but here we are humans releasing tons and tons of carbon into the atmosphere totally changing the the the set point as you described it like the homeostatic set point or whatever the non-equilibrium steady-state density that the atmosphere of the earth currently resides in and has resided in for x million years right so but we are changing that and we are seeing the earth adapt to us and the changes that we're making in it just in terms of hurricanes storms i mean the weather patterns are completely and and they're destroying large amounts of humans right and drought famine all of the things that are going to cascade through climate change as we see it progress so is this not adaptive active inference on the part of the earth or or what do you think of that really interesting i think in order to make that particular type of assumption one would actually have to you know ascribe to something like a guy hypothesis like lovelock sky hypothesis or the same type of thing that then margulis was into later in her life and i think that's that's that's an assumption that has to be made uh given the fact that okay well just because uh response happens chemically doesn't necessarily mean that it's not actually an adaptive response we you know uh gene expression is biochemical adaptation right so it's not about just being chemical uh using chemical or or something like this uh composition changing molecular structure as some type of uh criterion for adaptation but there's a i think there's an assumption that has to be made about what the system is there and if you're thinking about the earth as a maybe the earth is adapting to its constituent parts uh but that would involve that would involve uh intrinsically the earth being a particular type of living system given the fact that it has living parts one other quick thought on that is as a model comparison framework so go back to spm dcm all of our categorical model comparison techniques in the toolkit we can actually evaluate should we have cells as the units and then organisms then the each system in the world or should we use a different partitioning which one of these models is explaining predicting designing controlling better or which one is more parsimonious i prefer simplicity over utility or i have a function that balances simplicity and utility there's all kinds of frameworks that's kind of part of pluralism is starting with and being with multiplicity of answers and then knowing who we are and what we use to decide on which models to use so stephen you know i'm very much like to keep that ecological niche in the picture you know if it's organism in niche and groups of organisms in niche and then watershed a micro environment or micro climate as a niche i think that kind of becomes like a defining organism or unit in a way is like watch the watershed what's the kind of climate as experienced by that ecosystem and then of course you've got many ecosystems

around the world from the arctic to the antarctic you know and it will be in between now they might all be collectively affecting the broader carbon dioxide levels of the broader but when but how that then jumps to make the whole world an organism becomes for me slightly challenging right because it's it's always it was always a niche up to then and then suddenly it jumps to be an organism because we look at it from space and we see a nice round ball and it sort of looks alive so i wonder whether there's a so one one of the things i'm really interested in is like you're saying how things get ascribed the niche and we we talked about the path walking on the path and in a way like that dispersive thing animals can sort of also intuit some of that as well from the affordances and the directions and again bringing stuff in from systemic design type approaches is one thing that's interesting is if you then say humans like humans what we do is as a child and maybe some apes do this but not many is we point at something as a child and then we point to the parent and that's becomes it's in the niche but it becomes what they would call a boundary object in systemic design so you're trying to bring together a team around something which is beyond one discipline so what you need to do is you say okay we can't just use the variational free energy landscape of domain ecosystem which we have all attuned to in our sort of own way because now people are running by different rules right we're running by the physicists and the biologists right they're just going to be so you get a boundary object which you point to so it's something in the niche and then you sort of work to that um so anyway i just think there's something quite interesting then about like daniel was saying there about you know what sort of language and do we use but this scale problem comes back every time but i i wondered um like the difference between a boundary object or something which can be defined which i'm not sure if other animals can really do in the same way maybe a dog when it urinates up a tree i was just thinking about that does that become a boundary object in the in their world i don't know but um yeah anyways something that comes to mind thanks steven and with the transdisciplinarity in cancer biology they've recognized the the tumor and the stem cell niche and the micro environment and there's so much more to say on these very interesting topics just in the last few minutes we can go to a few of these final um points in the dot one it's always awesome to have the author and unpack some of the ideas and so this is kind of our little fractal.2 setting us up for our discussion next week and thinking about implications and how does it influence how we act moving forward and what kinds of questions we're looking at you wrote about some of the implications for thinking um individuality from a cognitive perspective you say it broadly it'd be awesome to hear your thoughts on the folks psychological notion of the cognizer and where

does the fep come into play thanks for the question uh so the folk psychological notion of again um my argument is a piggybacks upon this idea of the psychological biological notion of the individual and once we kind of free ourselves from thinking about uh individuals specifically in terms of replication dynamics uh that opens up a space for us to think about cognizers that are discontinuous with uh replication or replicators and i think this is like that that's going to be very important uh again when taking the uh this particular route of thinking about the evolution of cognition uh when thinking about for example okay if cognition goes all the way down to a certain to a certain degree then what part does it actually play in the the construction of larger systems uh and why would why would the construction of larger systems actually be some type of adaptation what uh for each of the systems involved when we're thinking about for example i think one particular thing that is quite important will be quite important to think about this for is okay the manner in which we think about artificial intelligence for example one way of course two ways of course of doing this uh is you know we go for various kind of specific tasks whether it's going to be kind of computer vision or uh gpt3 type tasks where we can actually analyze large corpus of text and kind of recapitulate that uh that's one thing the other thing of course is the task general or kind of domain general notion of ai and i think in order to perhaps tread into the department or turn into that field of more kind of task generality we're going to have to think about even simpler questions as to how can a system actually start maintaining itself uh according to its own tasks and again we come back to this notion of how can we actually uh set a system up you know to have its own intrinsic motivation and one way to do this is perhaps to piggyback on living systems so you introduce a living system to an artificial system and have to have them communicate in such a way uh that's you know analogous to some type of symbiotic system this is something again perhaps that mike levin michael levin has been thinking about doing actually in his lab uh and uh and i think this this is quite an interesting kind of method to think about the importance of uh of symbiosis in developing artificially intelligent systems you know um a system that can identify go go to go to a room an embodied intelligent agent that can like you know traverse a room and identify all the dogs that's great with it with an accuracy a high accuracy level that's fantastic but if if they can't respond to the dog knowing that a dog is going to actually you know dismantle it i'm not sure if i would want to call how intelligent that system would even be in terms of having basic intelligence and i think that fep and active influence allows allows a particular type of perspective on the notion of basic intelligence or intelligence that's surrounded or yes that's that's grounded in the notion of kind of uh a nest a nest density or

grounded in the notion of a type of a moving set point that's equivalent to a homeostatic moving homeostasis moving from static states and autostasis uh supporting states thank you matt shannon thanks for that i wonder um how important sort of drawing on what flu said and dave and steven i think also earlier some sort of effective or emotional state or awareness is in knowing and cognizing and um like how important that is for for a symbiotic relationship to be called a and whether you know like the neural network that is working with the human maybe to compose a piece of music um how important it is for that neural network to have cognizing capabilities of its own or maybe some sort of effective awareness or some sort of effective processing that it is able to develop in some way on its own or maybe only in symbiosis with a person and i don't know if if this bobtail squid and the vibrio like do they have sort of effective awareness together or does this would seem to you know experience something like pain i suppose would be one of the most basic effective states we might have so how important would basically is effective processing in fi delineating individuals or cognizers a great question shannon i'll take a go at it um i do think that at a certain level of complexity affective uh affective states play a huge role but i'm but because i'm i'm i suppose i'm making an assumption that cognition goes all the way down and not every system is going to be an affect have that capacity to have some type of effect actual awareness one might actually think about some other type of uh the sub-component plays a similar role as affect and perhaps that just might be something like um something that grounds that grounds a particular type of motivation to stay uh to stay in homostatic states or to remain in particular homeostatic states and this can be um oh pamela lyon has written about this just recently actually um it's not affect necessarily but it's valence negative and positive valence that might not and it has less to do with awareness or sentience in that sense uh but more to do with actually a a particular type of organism or particular type of system responding in ways uh that are responding in ways that keep it centered and whether or not this whether or not that kind of validates a bacteria might climb a gradient of glucose for example but bacterium might not have any type of uh you know experience of that glucose gradient but somehow we might say that that bacterium is actually it's it's positively valenced it's something that it's attracted to right uh and maybe in maybe in simpler systems you have something more like the notion of valence rather than affect but a certain period of uh you know complexity after a certain perhaps a degree of complexity arises affect is very important and i think particularly when it comes into multicellular systems perhaps uh that have to actually compute larger like much more information has to compete much more information and has to interact in their environments

their environmental niches are much more uh using a term by chem sterling much more con
opaque so uh effective information effective the matter in which we can use affect and how
that effect how that affects uh cognition i think it becomes quite important important but i'm
not exactly sure what what the number one is probably not a line uh it probably develops but
just what that function is is a different story i don't know do you have any ideas about that i
don't have an idea about where that would start but i feel like it would fall off so we might
have the small scale like our bobtail squid and vibrio who might have something like valence
that they're responding to and then as we get slightly larger organisms you know dogs people
then affect will play a lot more of a role but then as we have bigger systems like multiple
humans what is the effective like state of a crowd of people like we can ascribe like they look
angry or they sound angry but they as a crowd are they something as a crowd kind of like
group intentionality is there a group effective state it might kind of start falling off again so
you might have this sort of u-shaped curve of where valence matters and then somewhere here
in the sweet spot effective states or emotions really matter that's a that's super definitely
something to absolutely something i want to think about oh and also reminiscent of the
discussion of affect as well as higher order parameters in deep parametric active inference
where we even talked about which states were opaque or not whereas opaque usually meaning
like you can't see it you can't see in but actually the opaque mental states are the ones that we
do experience so that was in the sandved smith paper and whether other systems feel things
what is it like to be a squid what is it like to be a symbiosis that's the phenomenology and the
metaphysics but what is it like for us to model that system as if there's a higher level control
parameter we can know what that's like it's a lot like this conversation so just everybody can
provide a last comment on this awesome discussion i think the takeaway for me is that in a um
post-darwin biology it's easy to use the simplified trifecta of evolution by natural selection
phenotypic variability heritability of the phenotype fitness correlation and variation in fitness
and then project that onto all the kinds of individuality cognitive rights and responsibilities the
economic unit the sentient unit all of those as just simply under the coattails of the darwinian
replicator and i think pluralism and a bevy of other perspectives suggests that that's just not the
case blue and then anyone else with a last comment i just wanted to say thanks um for coming
i really enjoyed this discussion and i just started reading your paper your 2021 paper with
pizzullo uh the modeling ourselves um and i hope you'd consider coming back to talk about it
with us thanks so much it's been a really incredible conversation discussion so thank you all
for inviting me um shannon or stephen or dave any last comments awesome well all

participants thank you so much and matt especially for writing the paper and coming to talk about it and seeing the dot zero before so peace out everybody see you next week bye